

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 7: Weather, Climate and Adaptations of Animals to Climate

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	7 — Weather, Climate and Adaptations of Animals to Climate
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This hardest tier stacks multi-step traps: averaging an extreme into a climate verdict, confusing the value of an extreme with the clock-time it occurred, mismatching a weather element to its instrument, mixing depth-versus-volume rainfall logic, and running the surface-area-to-volume rule the wrong way for hot versus cold bodies — always decide weather-versus-climate FIRST, then compute, then refuse the look-right distractor. Do not write on this paper — mark answers on your answer sheet.

1. A town's 30-year January average maximum is 19 degrees C. This January, one freak day hit 31 degrees C, and a journalist wrote 'January is now 12 degrees hotter, the climate has flipped.' A student recomputes: if that single 31-degree day replaces a typical 19-degree day in the 30-year x 31-day January record, by roughly how much does the long-run January-maximum average actually move, and what does that show?

- (A) It rises by only about 12 divided by 930, i.e. roughly 0.01 degrees C, showing one hot day is swamped by hundreds of ordinary days and cannot by itself shift a multi-year climate average
- (B) It rises by the full 12 degrees C, because the hottest reading always sets the new climate average
- (C) It rises by 6 degrees C, the half-way point between the old average and the freak day's value
- (D) It cannot be estimated at all, because a single unusual day makes the whole 30-year record meaningless

2. Two reports about the same village arrive together: Report 1, 'the afternoon turned unexpectedly cold and a hailstorm struck for twenty minutes'; Report 2, 'in any given year roughly four months here are reliably below 10 degrees C.' A planner must decide which report to use to set the village's annual fuel-stocking schedule. The soundest decision is:

- (A) Use Report 1, because the hailstorm proves the village needs fuel immediately and so defines the yearly plan
- (B) Use Report 2, because a yearly fuel schedule must rest on the repeating long-term climate pattern, not on one afternoon's unusual weather
- (C) Use neither, because fuel planning depends only on the price of fuel, not on temperature at all
- (D) Average the two reports together, since combining a single day with a yearly pattern gives the most balanced plan

3. A child says, 'Last July it rained on 28 of 31 days, but this July it rained on only 6 days, so our region's climate is drying out for good.' A teacher calls this a classic confusion. The precise error is that the child:

- (A) Forgot that rainfall is never part of climate and belongs only to weather, so the comparison was meaningless
- (B) Assumed July must always be the wettest month, when in fact no month is ever wetter than another
- (C) Compared two single years of weather and read the gap as a permanent climate change, when climate is judged only from a pattern averaged over many years
- (D) Used the number of rainy days instead of the total millimetres, which is the only correct way to judge climate

4. City S's last six annual mean temperatures were 25.9, 26.1, 26.0, 25.8, 26.2, 26.0 degrees C; City T's were 9, 24, 12, 27, 8, 26 degrees C. A reporter writes 'both average about 18 to 19 degrees, so their climates are basically the same.' The most accurate climatic correction is:

- (A) The reporter is right, since two places whose multi-year mean temperatures coincide must share one climate
- (B) City T has no climate at all, because a place whose yearly figures jump around cannot be said to have one
- (C) The averages alone hide that S is steady year to year while T swings violently, so their climates differ sharply even where the means happen to land near each other
- (D) City S has no weather, since temperatures that barely change from year to year mean nothing is happening there

5. A geography quiz asks which single pairing is correct: 'It is foggy in Shimla this morning' versus 'Shimla has cool summers and snowy winters every year.' Identify which is weather, which is climate, and whether both can simultaneously be true of Shimla:

- (A) The foggy morning is weather and the recurring cool-summer-snowy-winter pattern is climate, and both can be true of Shimla at the same time
- (B) Both statements describe climate, because each one names Shimla and a temperature-related condition
- (C) Both statements describe weather, because each one mentions something you could see or feel outdoors
- (D) They cannot both be true, since a place with snowy winters cannot also have fog on a given morning

6. A region's 20-year record shows its mean temperature unchanged, yet within those 20 years lie both the hottest day and the coldest night ever logged. A debate club argues over what this proves. The single correct reading is:

- (A) Record-breaking single days are extreme weather events that can sit inside an unchanged climate, because the long-run average is not moved by isolated extremes
- (B) The climate must be becoming unstable, since holding both a record-hot and a record-cold day is impossible for a stable climate
- (C) The hottest day and coldest night must exactly cancel, which is the only reason the 20-year mean stayed level
- (D) One of the two records must be a faulty reading, because a steady average cannot coexist with genuine extremes

7. Two questions are posed: (i) 'Will I need a raincoat for my walk in the next half hour?' and (ii) 'Which months should our town reserve its disaster fund for floods?' Match each to the data it truly needs:

- (A) Both need the long-term climate record, since any sensible decision should be based on many years of data
- (B) Both need only the present forecast, because both decisions are being made today using current information
- (C) (i) needs the next half hour's weather forecast, while (ii) needs the town's long-term climate of which months flood year after year
- (D) (i) needs the climate record and (ii) needs the immediate forecast, since floods can begin within the next half hour

8. A travel writer reasons: 'A desert is defined by no rain, yet I saw it rain there, so it cannot really be a desert.' A geographer untangles two ideas. The exact mistake is that the writer:

- (A) Was correct, because any place that ever receives rain automatically fails to qualify as a desert
- (B) Confused a single rainy spell of weather with the desert's long-term climate, which is defined by scarce rainfall averaged over years, not by every day being dry

- (C) Forgot that deserts are defined by their high daytime temperature alone, so rainfall is irrelevant to the label
- (D) Assumed rainfall belongs only to climate and never to weather, which is what made the trip misleading

9. A scientist plots City U's monthly mean temperature for ten straight years: every year the curve climbs from a January low to a July high and falls back, the same shape repeating. On the same page she plots City V's hour-by-hour temperature for one day, a jagged line with no repeat. Each plot is best labelled:

- (A) Both plots show climate, because both were drawn from real measured temperatures of a place
- (B) The repeating monthly curve shows weather and the one-day line shows climate, because the monthly plot covers more dates
- (C) The ten-year repeating monthly curve shows climate, while the one-day jagged hourly line shows weather
- (D) Both plots show weather, because temperature is a weather element no matter how long it is recorded

10. A student claims a place's climate has 'no day-to-day weather' because its yearly average temperature has stayed near 27 degrees C for fifteen years. The cleanest correction is:

- (A) A steady long-term average describes a stable climate, but the place still has daily weather that rises and falls around that average from day to day
- (B) The student is right, because a constant yearly average means each day must have exactly the same weather
- (C) The student is right, because climate replaces weather entirely once an average has been calculated
- (D) The student is wrong because a stable average actually proves the climate is changing rapidly

11. A five-day log reads — Max (deg C): Mon 29, Tue 26, Wed 37, Thu 34, Fri 30; Min (deg C): Mon 18, Tue 19, Wed 16, Thu 28, Fri 20. A report must name the day with the SMALLEST daily temperature range. After computing each day's range, the correct day and value are:

- (A) Wednesday, with a range of 21 degrees C, because Wednesday had the highest maximum of the week
- (B) Tuesday, with a range of 7 degrees C, because Tuesday had the lowest maximum of the week
- (C) Thursday, with a range of 6 degrees C, the smallest of the five days
- (D) Monday, with a range of 11 degrees C, because Monday's readings were the earliest pair of the week

12. Using a week's rainfall (mm): Mon 0, Tue 8, Wed 0, Thu 0, Fri 12, Sat 36, Sun 4, a report needs the number of rainy days, the weekly total, and the average rainfall PER RAINY DAY. The correct set is:

- (A) 7 rainy days, total 60 mm, average about 8.6 mm per day, dividing by all seven days
- (B) 4 rainy days, total 60 mm, average about 8.6 mm, dividing the total by all seven days anyway
- (C) 3 rainy days, total 56 mm, average about 18.7 mm, counting only the three heaviest days
- (D) 4 rainy days, total 60 mm, average 15 mm per rainy day

13. Over four days a city's maxima were 27, 31, 36 and 30 deg C and its minima were 17, 19, 22 and 18 deg C. Which value is closest to the AVERAGE DAILY temperature RANGE across the four days?

- (A) 12 degrees C, found by averaging the four daily ranges 10, 12, 14 and 12
- (B) 31 degrees C, the average of the four daily maximum temperatures
- (C) 19 degrees C, the highest maximum minus the lowest minimum across the four days
- (D) 9 degrees C, the average maximum minus the average minimum, but using only the first two days

14. A max-min thermometer is read at 8 p.m. and shows max 34 deg C, min 21 deg C. The observer was asleep all afternoon and saw nothing, yet trusts the figures. A sceptic asks how max and min can be known with no one watching. The correct account of the instrument is:

- (A) Small index markers are pushed to the highest and lowest points reached during the day and stay there, so the extremes are recorded even with nobody watching
- (B) It continuously shows only the temperature at the exact instant it is read, which the observer assumes was also the day's extreme
- (C) It samples the temperature once each hour and the observer averages those samples to get the max and the min
- (D) It measures the rainfall and the temperature together, and the rainfall figure is converted into the day's maximum

15. A rain gauge's collecting jar is only 120 mm deep. After a record storm the jar is brimful and water is spilling over the rim; the observer logs '120 mm'. The key data-quality lesson is:

- (A) The rainfall was exactly 120 mm, since that is the highest number the full jar can display
- (B) The rainfall was at most 120 mm, because a jar cannot collect more water than its own depth allows

(C) The overflow means rain stopped the instant the jar filled, so 120 mm is the precise final figure

(D) The true rainfall was at least 120 mm; an overflowing gauge can only establish a lower bound, never the exact total

16. A bulletin lists: temperature 30 deg C, humidity 68%, rainfall 9 mm, wind 24 km/h, sunshine 6 hours. A teacher asks which figure was obtained from a hygrometer AND which separate figure needs an anemometer. The correct pair is:

(A) The 68% came from the hygrometer and the 24 km/h came from the anemometer, each a distinct instrument for a distinct element

(B) The 30 deg C came from the hygrometer and the 6 hours from the anemometer, since both react to the Sun

(C) The 9 mm came from the hygrometer and the 68% from the anemometer, since both involve water in the air

(D) The 24 km/h came from the hygrometer and the 68% from the anemometer, swapping the two instruments

17. A daily humidity log reads: 05:00 -> 92%, 10:00 -> 74%, 14:00 -> 52%, 18:00 -> 67%, 22:00 -> 88%. The dip to its lowest at 14:00 is best explained by:

(A) The air genuinely held the least water vapour of the entire day precisely at 14:00 because vapour escapes upward at that hour

(B) The hygrometer automatically resets itself to a low value in the early afternoon every day

(C) Early-afternoon sunshine has warmed the air toward its hottest, and warmer air reads a lower relative humidity for the same amount of moisture present

(D) The wind was strongest at 14:00 and physically blew the moisture out of the instrument's chamber

18. Rainfall is reported as a DEPTH in millimetres rather than as the LITRES a gauge collects. The deepest reason this lets distant stations be compared fairly is:

(A) Depth measures how fast the rain fell, and rainfall speed is the only fair basis for comparing storms

(B) Litres cannot be measured by any rain gauge, so depth is simply the only quantity that remains available

(C) Depth does not depend on the width of the gauge's mouth, so the same storm yields the same depth on a narrow gauge and a wide one, while litres collected would differ with mouth size

(D) A wide gauge always collects fewer litres than a narrow one, which would make litres a misleading figure

19. A station logs hourly temperature and finds the air hottest near 2 p.m. though the Sun is highest at noon. On the same graph, sunlight intensity peaks at noon while the temperature curve peaks two hours later. The lag is best explained by:

- (A) The Sun is physically closest to the Earth at 2 p.m. rather than at noon, delivering its strongest rays then
- (B) The thermometer reacts slowly and needs about two hours to catch up with the real air temperature
- (C) Humidity drops to zero at 2 p.m., and it is the perfectly dry air that makes the early afternoon hottest
- (D) For a couple of hours after noon the ground still gains heat faster than it loses it, so the air keeps warming even as the Sun begins to sink

20. An observer records temperature all night and finds it lowest just before sunrise, not at midnight. The best Sun-based explanation is:

- (A) Midnight is the instant the Earth is farthest from the Sun each day, so it ought to be the coldest moment
- (B) With no incoming sunlight the ground keeps losing stored heat all night, so the air is coldest after the longest stretch of cooling, just before the Sun returns
- (C) The Moon actively pulls heat from the ground until midnight, after which the air begins to warm again
- (D) The thermometer drifts steadily lower overnight by itself, regardless of how much heat the ground has lost

21. Two spots a few metres apart are read at the same clock-time on a clear afternoon: full sun reads 39 deg C, deep shade reads 30 deg C. A student must state the single firmest conclusion. It is:

- (A) It is the incoming sunlight, not the time on the clock, that does the heating, since equal-time spots differ only by where the sunlight falls
- (B) The clock itself controls temperature, since both readings were taken at exactly the same time
- (C) Shade is colder only because cold air sinks and pools into shaded places while avoiding sunlit ones
- (D) The shaded thermometer must be faulty, because two thermometers at the same time should always read alike

22. Why is the very same slanting-rays principle that keeps the poles cold also the reason a winter noon feels weaker than a summer noon in one mid-latitude city?

- (A) In winter the Sun moves much farther from the Earth, so every ray it sends arrives weakened
- (B) In winter the Sun stays lower in the sky, so its rays strike more slantingly and spread the same energy over more ground, exactly as polar rays do year-round
- (C) In winter the Sun actually shines for more hours, but each hour somehow carries far less heat
- (D) Winter air is physically thicker and simply blocks most of the sunlight before it can reach the ground

23. Sunlight is the ultimate driver of weather. Which chain best captures HOW its energy produces both wind and rain?

- (A) Sunlight heats land and sea unevenly, the warmed air rises and sets up pressure differences that drive wind, while the same heat evaporates water that later condenses into clouds and falls as rain
- (B) The Sun's rays physically push on the air with their pressure, shoving it sideways to create wind directly
- (C) The Sun cools the oceans so the chilled water can then rise as vapour and fall again as rain
- (D) Moonlight by night and sunlight by day share the work equally, the Moon driving the wind and the Sun the rain

24. A region near the equator and a region near a pole each receive sunlight for a full year. Which statement about the SUN'S ENERGY per unit area correctly explains the climate contrast?

- (A) Both regions receive the same energy per unit area, so any climate difference must come from wind alone
- (B) The poles receive more total sunlight because their summer days are very long, so they should be the warmer of the two
- (C) The equator is cold and the poles are hot, because slanting rays carry more heat per area than direct ones
- (D) The equator receives intense, nearly overhead rays concentrating energy on a small area, giving a hot climate, while the poles receive weak slanting rays spread thin, giving a cold climate

25. A meteorologist forecasts tomorrow well but cannot name the weather on this exact date next year, yet she CAN describe next year's likely season. Why is this consistent rather than contradictory?

- (A) The Sun becomes too weak a full year ahead to make any weather, so only the broad seasons remain knowable
- (B) Tiny differences among the many interacting atmospheric factors grow rapidly, blurring far-off daily detail, while the multi-year averaged seasonal pattern that is climate stays predictable
- (C) Climate is fundamentally impossible to study, so only the next day's short-range weather can ever be predicted
- (D) Thermometers stop working after a single day, which is exactly why long-range daily forecasts always fail

26. Four cities report (warmest-month mean, coolest-month mean, yearly rainfall): A (29, 27, 2400 mm); B (46, 7, 90 mm); C (3, -29, 150 mm); D (34, 17, 650 mm). Which city's data most clearly fits a HOT-DESERT climate?

- (A) City A, because its warmest month is hot and it receives by far the most rainfall of the four
- (B) City C, because its tiny rainfall resembles a desert even though both its temperatures are well below freezing-cold
- (C) City B, because it is very hot in its warm month, swings widely to a cool month, and receives almost no rain
- (D) City D, because it is warm and has only moderate rainfall, which is typical of dry regions

27. A habitat card reads: temperature 24-29 deg C every month, humidity above 85%, rainfall over 2300 mm spread through the year, dense tall canopy; the resident animal has grasping hands, a long tail and loud calls. The single best joined-up explanation is:

- (A) The figures describe a polar region, where a grasping tail grips the ice and loud calls echo across open snow
- (B) The constant warmth, heavy year-round rain and tall trees define a rainforest, where grasping limbs aid climbing and loud calls carry through dense foliage that blocks sight
- (C) The figures describe a hot desert, where a long tail stores water and loud calls travel far over open sand
- (D) The figures describe a cold mountain snowfield, where strong limbs dig through snow and the calls warn of cold

28. A nature note states: 'A thick fat layer, a compact rounded body and white fur all serve one common climatic purpose.' Name the shared purpose, and predict the fate of such an animal in a tropical rainforest:

- (A) All three conserve heat or hide the animal in snow, so in a hot, wet rainforest it would overheat and stand out conspicuously against the green leaves
- (B) All three help the animal lose heat quickly, so it would stay comfortably cool throughout the rainforest
- (C) All three are for camouflage among green leaves, so it would be perfectly hidden in the rainforest
- (D) All three store drinking water, so it would survive the rainforest easily by drawing on its reserves

29. Two penguins differ only in shape: X is compact and rounded; Y is long, slender and has a large thin tail. In identical freezing Antarctic conditions, the likely outcome is:

- (A) Y survives better because its larger surface area soaks up more of the Sun's warmth in the cold
- (B) Both survive equally well, because an animal's body shape has no effect on how fast it loses heat

- (C) X conserves heat better because its lower surface-area-to-volume ratio loses heat more slowly, while Y's larger exposed surface loses heat too fast
- (D) X loses heat faster, because a compact body actually exposes more surface to the surrounding cold

30. A polar bear has dense water-shedding fur over thick fat, plus wide paws and strong swimming. Which statement pairs EACH feature with its correct main job?

- (A) The fat layer's main job is to make the bear float higher than its prey so it can ambush from above
- (B) The dense fur's main job is to keep the bear cool during the brief polar summer days
- (C) The swimming ability's main job is to let the bear climb trees once the sea ice melts away
- (D) Wide paws spread weight on snow and act as paddles in water, while the fur-plus-fat combination slows heat loss in cold air and icy water

31. A reviewer writes, 'The polar bear is white so that it absorbs the most sunlight and stays warm.' The correct response is that the white coat is:

- (A) Correct, because white surfaces absorb the greatest amount of heat from sunlight
- (B) White only so that other polar bears can spot it easily from far across the ice
- (C) White so that its prey can see it approaching, making the hunt fair for both sides
- (D) Mainly camouflage that blends the bear into snow and ice while it hunts, not a device for absorbing the Sun's heat

32. Penguins press into a tight huddle during the coldest, windiest spells. The correct cause-and-effect account is that huddling:

- (A) Increases each bird's exposed surface, which helps the crowded group shed dangerous excess heat
- (B) Lets the birds catch fish more easily, because crowding on the ice herds the fish toward them
- (C) Reduces the total body surface each bird exposes to the cold air, so the group as a whole loses heat more slowly
- (D) Traps a pocket of cold air between the birds, which cools them and thereby saves energy

33. An elephant in a hot region has very large, thin, blood-rich ears that it flaps; an arctic fox has small, well-furred ears. The single principle explaining BOTH ear designs is:

- (A) Both ear types are mainly for hearing, the large ones catching distant sounds and the small ones hearing under snow
- (B) Large ears trap heat to keep the elephant warm, and small ears trap heat to keep the fox warm too

- (C) Large thin ears expose much surface to shed excess heat in the warmth, while small ears expose little surface to conserve heat in the cold
- (D) Ear size is fixed by body size alone and has no connection to the climate an animal lives in

34. Bright-green rainforest frogs hide among leaves while polar bears are white against snow. The deep idea shared by both colour schemes is that a colour is favoured when it:

- (A) Is whichever colour best warms the body, green absorbing heat in forests and white absorbing it in snow
- (B) Glows in the dark, helping the animal be seen by others of its own kind in any habitat
- (C) Is the brightest possible colour, so that predators are frightened off by the boldness
- (D) Matches the usual background of the animal's habitat, hiding it from predators or prey in that climate's surroundings

35. A monkey with grasping hands, a long balancing tail and loud calls is moved to open polar ice. It would most likely struggle because every one of these features suits:

- (A) Surviving intense cold, so the only real problem would be getting used to the unfamiliar terrain
- (B) Burrowing through hot dry sand, which is precisely why the freezing ice would defeat it
- (C) Moving and communicating in a warm, wet, densely-treed forest, none of which exists on bare polar ice
- (D) Flying long distances over the open ocean, which the solid ice unfortunately prevents

36. Why do tropical rainforests support a far greater VARIETY of animals than polar regions? The best causal explanation is:

- (A) Polar cold makes animals reproduce much faster, yet the rainforest still wins simply because it is larger
- (B) Rainforests have almost no plants, so animals there face no competition and so multiply freely
- (C) Variety depends mainly on how many human visitors a region receives over the course of a year
- (D) The warm, wet, stable rainforest climate provides abundant and varied food and shelter all year, letting many different kinds of animal coexist

37. An animal is described only as 'a long, slender body with large, thin, well-exposed ears and short fur.' Without naming a species, the climate this body plan best suits, and why, is:

- (A) A polar climate, because large ears catch and hold extra warmth against the surrounding cold
- (B) A hot climate, because the large exposed surface and thin fur together help the animal shed body heat rather than trap it

- (C) A polar climate, because a slender body conserves heat far better than a compact rounded one
- (D) Any climate equally, because body shape and ear size make no difference to how heat is lost

38. Two stations at the SAME latitude are compared over a full year: coastal C has a small annual temperature range and high humidity; inland I has a large range and lower humidity. Since latitude is equal, the best explanation is:

- (A) Station C must lie at a far higher altitude, which is what shrinks its annual temperature range
- (B) The nearby sea moderates station C's temperatures and adds moisture, while inland station I, far from the sea, swings more widely and stays drier
- (C) Station I receives more direct sunlight than C because inland places sit physically closer to the Sun
- (D) The two differ only in the particular days on which each happened to be measured, not in any lasting way

39. A six-day sunshine record (hours) reads: Mon 9, Tue 1, Wed 8, Thu 0, Fri 7, Sat 2. A student must say what this best illustrates. The soundest reading is:

- (A) Sunshine swung sharply from day to day, illustrating how quickly weather can change over a short period
- (B) The figures prove the region's whole climate changed during this single week
- (C) Sunshine was effectively constant across the week, illustrating a very stable climate
- (D) The Sun shone for the same number of hours each day regardless of the clouds that were present

40. A region records every year: summer maxima around 48 deg C, winter minima around 4 deg C, almost no rain, and very wide day-to-night swings. The correct combined reading of these climate signatures is:

- (A) A tropical rainforest, because the high summer maxima show the place is hot and therefore must be wet
- (B) A polar climate, because such wide temperature swings closely resemble the extremes found near the poles
- (C) A temperate coastal climate, because the winter minimum staying above freezing rules out anything harsher
- (D) A hot-desert climate, where clear dry air heats strongly by day and loses heat fast at night, giving large swings and very little rain

41. Two adaptations are (i) a thick layer of fat under the skin and (ii) long limbs with a grasping tail. Each helps in one climate and hurts in the other. The correct assignment of both is:

- (A) Both features help in the cold, the fat for warmth and the grasping tail for gripping the ice
- (B) Both features help in the rainforest, the fat for cushioning against the heavy rain and the tail for climbing
- (C) Neither feature depends on climate, since fat and limbs serve the very same purpose everywhere
- (D) The fat layer aids a cold-climate animal but would cause overheating in a hot rainforest, while grasping limbs aid a rainforest climber but are useless on bare polar ice

42. A naturalist compares a compact, rounded polar animal with a long, lean, big-eared desert animal of a related kind. Asked for the ONE underlying rule, the best answer is:

- (A) Compact bodies lose heat faster, which is what the cold demands, while lean bodies lose heat slower, which suits the heat
- (B) Body shape only changes how large each animal looks to its predators and has nothing to do with heat
- (C) Surface-area-to-volume ratio governs heat exchange: a compact body has little surface for its volume and conserves heat in the cold, while a lean body with big ears has much surface and sheds heat in the warmth
- (D) The desert animal is lean because its food is scarce, while the polar animal is round because food is plentiful

43. A Siberian crane breeds in northern Russia and appears at Indian wetlands only in winter, vanishing every Indian summer. Which statement correctly separates the role of CLIMATE from day-to-day weather in this migration?

- (A) The crane reacts to the particular weather of the one day it happens to set out, choosing only sunny days to fly
- (B) The crane follows the regular seasonal climate cycle, leaving the harsh northern winter and returning when the north warms, not reacting to any single day's weather
- (C) The crane dies out every Indian summer and a fresh set of birds simply appears each winter
- (D) The crane transforms into a heat-loving species for the Indian summer, a change triggered by hot weather

44. A flock breeds at a northern wetland each spring and flies south each autumn for many years. What governs this regular back-and-forth, and what cue most likely starts the autumn departure?

- (A) The random weather of a single chosen day governs it, and a sudden cold snap on that one day is the cue
- (B) The birds permanently settling governs it, and the cue is simply that they stop wanting to fly any further

- (C) A one-time event that never repeats governs it, and the cue is the changing colour of one particular tree's leaves
- (D) The regular seasonal climate cycle governs it, and shortening day length together with a falling food supply is the most likely cue that starts the southward flight before the worst cold arrives

45. A rain gauge at Station A has a wide funnel; an identical-depth gauge at Station B has a narrow funnel. The same rainstorm passes over both. What will their readings show, and what does this reveal about why depth is used?

- (A) The wide gauge reads a greater depth, because it collects more water overall during the storm
- (B) The narrow gauge reads a greater depth, because the same water is squeezed into a thinner column
- (C) The two readings cannot be compared at all, because their funnel sizes differ
- (D) Both read the same depth, because depth is independent of funnel width, which is exactly why depth lets different stations be compared fairly

46. Why can a meteorologist say 'next July will probably be the wettest month here' yet be unable to say whether it will rain on the 14th of next July? The single principle that resolves the puzzle is:

- (A) The Sun behaves predictably across all of July but unpredictably on individual days, so only whole months can be forecast
- (B) Rainfall obeys the calendar, so months are fixed in advance but individual dates are chosen at random by the weather
- (C) The long-term averaged climate, such as which month is usually wettest, is predictable, but the exact weather of one distant day is not, because tiny atmospheric differences grow rapidly over time
- (D) Thermometers and rain gauges work reliably for a whole month but fail the moment they are asked about one future day

47. A six-day record gives daily maxima for Place 1: 26, 27, 25, 27, 26, 27 deg C, and Place 2: 43, 18, 45, 16, 42, 17 deg C; the six-day averages happen to be similar. The most accurate climatic statement is:

- (A) The two places have identical climates, because their six-day average maxima come out close to each other
- (B) Place 2 has no climate at all, because a place whose temperature jumps so much cannot be said to have one
- (C) Place 1's steadiness proves it has no weather, since nothing about it ever changes from day to day
- (D) Place 1 has small day-to-day swings and Place 2 has very large ones, a difference in their climates that the similar averages hide

48. A weather page reports for one city: temperature 32 deg C, humidity 71%, rainfall 0 mm, wind 26 km/h, sunshine 7 hours. A reader wants the ONE figure telling how much water vapour the air holds, and why the others mislead. The correct choice is:

- (A) The 0 mm rainfall, because rainfall directly counts the water that is currently in the air
- (B) The 71% humidity, because humidity is the measure of water vapour in the air, expressed as a percentage
- (C) The 32 deg C temperature, because hotter air always means more water vapour must be present
- (D) The 26 km/h wind speed, because moving air carries the moisture and so directly measures it

49. A weather summary reads: 'Maximum 4 deg C, minimum -16 deg C, light snow, 19 hours of darkness.' A second reads: 'Maximum 31 deg C, minimum 26 deg C, heavy rain, dense forest nearby.' Matching each to a climate type, the correct pairing is:

- (A) The first points to a hot desert because its temperature range is wide, and the second to a polar climate because it has precipitation
- (B) The first points to a polar-type climate and the second to a tropical-rainforest climate
- (C) Both point to tropical climates, because both reports mention some form of precipitation
- (D) The first points to a temperate coast because its maximum is above zero, and the second to a desert because it is hot

50. A child argues, 'It is freezing in our town today, so we must live in a polar region.' The clearest reason this is mistaken is:

- (A) One freezing day is weather and can happen in many places, while a polar region is defined by cold persisting through most of the year
- (B) Cold days simply never occur anywhere except in genuine polar regions
- (C) A single cold day really does prove a polar climate, so the child is actually correct
- (D) Temperature plays no part at all in defining whether a region counts as polar

51. A red-eyed tree frog sits motionless and green on a leaf by day and is hard to spot. The strongest evidence that its colour is a genuine ADAPTATION rather than a lucky coincidence is that:

- (A) The colour reliably matches the green leaves of its rainforest home generation after generation, reducing the chance predators detect it
- (B) The colour keeps the frog warm during cold nights, which would be useful in any climate at all
- (C) The colour lets the frog reflect sunlight and stay cool, just as a hot-desert animal would need
- (D) The colour allows the frog to glow and lure prey toward it in complete darkness

52. An expert defines an adaptation as 'a heritable feature that, over long times, suits an animal to its climate.' Which observation best fits this definition?

- (A) Polar bears born generation after generation with thick fur and a fat layer, matching the long-cold polar climate
- (B) A single bear that happened to feel warm on one unusually mild afternoon
- (C) A monkey that learns to open a new kind of fruit during a single morning
- (D) A bird that flies indoors to escape a cold night on one particular occasion

53. A toucan has a very large but light beak, a hummingbird a slender one, and a polar bear thick fur. Match each to its right reason. The fully correct matching is:

- (A) The toucan's beak stores body fat for a cold season, while the bear's fur keeps it cool through the summer
- (B) The toucan's beak digs burrows in dry soil, while the bear's fur camouflages it among green leaves
- (C) The toucan's beak warms its head on freezing nights, while the bear's fur helps it swim faster through water
- (D) The toucan's light beak reaches fruit on thin branches it could not perch on, while the bear's thick fur insulates it against cold

54. A region's monthly mean rainfall (mm) for one year is: Jan 5, Feb 6, Mar 4, Apr 7, May 9, Jun 220, Jul 260, Aug 240, Sep 180, Oct 30, Nov 8, Dec 6. The climate this best illustrates, judged from the pattern, is:

- (A) A tropical-rainforest climate, because the total rainfall over the year adds up to a very large number
- (B) A strongly seasonal climate with a concentrated rainy season in a few months and a long dry stretch through the rest of the year
- (C) A polar climate, because most months receive only a small amount of precipitation
- (D) A climate with rain evenly spread through the year, because every single month records at least some rain

55. A reader sees three numbers from one city: a maximum of 33 deg C, a minimum of 19 deg C, and a temperature range. Asked which single quantity tells how much the temperature VARIED within that day, and to give its value, the correct answer is:

- (A) The temperature range, equal to 14 degrees C, since the range is the maximum minus the minimum
- (B) The maximum of 33 degrees C, because the highest figure shows how much the temperature varied
- (C) The minimum of 19 degrees C, because the lowest figure measures the day's variation
- (D) The sum of 52 degrees C, since adding the maximum and minimum gives the total variation

56. A student combines three facts: (1) the Sun heats the equator more strongly than the poles; (2) warm air rises and cool air sinks; (3) rainforests cluster near the equator while ice caps lie at the poles. The best single conclusion tying these together is:

- (A) The poles are hot because warm air sinks there, which is why ice can only form at the equator
- (B) The equator is cold because rising warm air leaves it empty of heat, so rainforests actually need the cold
- (C) Stronger, more direct equatorial sunlight makes that zone hot and supports wet rainforest climates, while weak slanting polar sunlight leaves those zones cold and icy
- (D) Sunlight strength has no real link to where rainforests and ice caps occur; their positions are accidental

57. A graph shows daytime air temperature rising from sunrise, peaking in the early afternoon, then falling toward sunset, while a second curve on the same axes shows the Sun's height peaking exactly at noon. The temperature peak LAGGING the noon Sun peak is best explained by:

- (A) The Sun delivers its greatest heat in mid-afternoon because it is then directly overhead
- (B) Even after the Sun starts descending from noon, the ground keeps gaining heat faster than it loses it for a couple of hours, so the air keeps warming past noon
- (C) The air takes a fixed two hours to physically travel down from the Sun to the ground
- (D) Clouds always gather at noon and block the Sun, delaying the heating until after they clear

58. On a still, sunny morning a shallow puddle on hot tar dries up within an hour, while an identical puddle in cool shade nearby is still wet. Linking this to how the Sun drives weather, the best explanation is:

- (A) The shaded puddle's water sank into the ground while the sunlit puddle's water was pushed sideways by the Sun's rays
- (B) The Sun's heat speeds evaporation, turning the sunlit puddle's water into vapour faster, which is the same process that lifts water into the air to later form clouds and rain
- (C) The sunlit puddle dried because sunlight is heavier than shade and physically presses the water out of the tar
- (D) Both puddles lost the same amount of water, and only the colour of the tar made one look drier than the other

59. Two coastal towns at the same latitude report their yearly humidity: town K averages 80% with a small annual temperature range, town L averages 45% with a large annual range, and L lies far inland behind a mountain. A student must reconcile the labels 'both coastal' with the data. The soundest conclusion is:

- (A) Both can be coastal, because humidity and temperature range have no connection to nearness to the sea
- (B) The data fit only if town L is not truly coastal; its low humidity and wide temperature range are the signatures of an inland location far from the moderating, moisture-supplying sea
- (C) Both can be coastal, and town L is simply drier because inland mountains always add moisture to the air
- (D) The labels are correct and the sea makes L drier, because the ocean removes water vapour from nearby air

60. A student writes a final synthesis. Which single statement is fully correct about weather, climate and adaptation together?

- (A) Climate is the short-term state and weather is the many-year average, and adaptations respond to each day's weather
- (B) Weather and climate are the same single measurement, and adaptations are habits an animal learns within one day
- (C) Weather is measured by instruments but climate cannot be measured in any way, and adaptations have nothing to do with climate
- (D) Weather is the short-term state of the atmosphere at a place, climate is the average of that weather over many years, and animals' adaptations are heritable features matching the long-term climate, not the weather of one day

Solutions — Science (Class 7), Chapter 7: Weather, Climate and Adaptations of Animals to Climate — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	C	21	A	31	D	41	D	51	A
2	B	12	D	22	B	32	C	42	C	52	A
3	C	13	A	23	A	33	C	43	B	53	D
4	C	14	A	24	D	34	D	44	D	54	B
5	A	15	D	25	B	35	C	45	D	55	A
6	A	16	A	26	C	36	D	46	C	56	C
7	C	17	C	27	B	37	B	47	D	57	B
8	B	18	C	28	A	38	B	48	B	58	B
9	C	19	D	29	C	39	A	49	B	59	B
10	A	20	B	30	D	40	D	50	A	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) A 30-year January holds about $30 \times 31 = 930$ daily maxima. Swapping one 19-degree day for a 31-degree day adds 12 degrees spread over 930 values, so the mean moves by about $12/930$, near 0.01 degrees C — essentially nothing. That is exactly why climate, an average over many years, barely twitches at one freak day; treating the extreme as the new average, or splitting the difference, both ignore the huge denominator, and the record stays fully usable, refuting the 'meaningless' claim.

2. (B) Stocking fuel for a whole year is a long-term decision, so it must track the recurring climate (Report 2's reliable four cold months), not a one-off weather event. A twenty-minute hailstorm is weather and cannot anchor a yearly plan; cost matters operationally but the question is which climate-or-weather signal to use; and averaging a single afternoon into a multi-month climate pattern mixes incompatible time-scales and dilutes the real seasonal signal.

3. (C) Two consecutive Julys are two samples of weather; a real climate trend needs many years averaged together, so one wet July and one dry July prove nothing permanent. Rainfall genuinely is a climate element when averaged, so that distractor is false; whether July is wettest varies by region and is not the error here; and counting rainy days is a perfectly valid summary — the flaw is the leap from two years to 'for good,' not the choice of measure.

4. (C) Wait — S averages near 26, not 18; only T's wildly scattered values (8 to 27) average near 18, and equal-looking means can mask opposite climates: S is remarkably stable, T is highly variable. So a shared mean does not imply a shared climate. T's variability is itself a climate feature, not an absence of climate; and S still has daily weather despite its steady yearly mean — a stable climate is not a dead one.

5. (A) A single morning's fog is the day-to-day state — weather; a pattern repeating 'every year' is the multi-year average — climate. The two describe different time-scales of the same place and are fully compatible. Naming the place does not make a statement climate; not everything observable outdoors is mere weather (the yearly pattern is climate); and fog on one morning sits comfortably inside a snowy-winter climate, so there is no contradiction.

6. (A) Extreme single days are weather; a 20-year mean is climate, and the mean can stay flat while individual days roam far from it. There is no need for the two records to 'cancel' — hundreds of ordinary days hold the average steady regardless of two outliers; a flat mean does not make a climate unstable, it shows the opposite; and genuine extremes are normal weather, not necessarily instrument errors.

7. (C) An imminent personal choice tracks short-term weather (the half-hour forecast), while a recurring yearly budgeting choice tracks the repeating climate (which months usually flood). Not every decision is long-term, so (i) is not climate; not every decision is immediate, so (ii) is not the present forecast; and reserving a yearly fund is inherently about the seasonal pattern, not whether rain starts in the next thirty minutes.

8. (B) A desert's defining feature is low rainfall on the multi-year climate average; an occasional rainy day is weather and does not disqualify the climate. Deserts can and do receive rare rain, so 'ever receives rain' is wrong; low rainfall, not temperature alone, defines a desert (cold deserts exist); and rainfall is part of weather day-to-day and climate when averaged, so that final option misstates the concept.

9. (C) A pattern averaged and repeated over many years is climate; a single day's minute-to-minute changes are weather. Being measured does not make data climate — the time-span and averaging do. Covering more dates does not flip weather into climate by itself unless it is averaged over years, but here the long repeating curve IS the climate and the single day is weather, so the swap is wrong. And although temperature is a weather element, a ten-year averaged seasonal cycle is a climate description.

10. (A) Climate is the long-term average; weather is the daily variation that still occurs around it. A flat fifteen-year mean does not freeze every day at 27 — individual days swing above and below. Climate does not 'replace' weather; the two describe different time-scales of the same place. And a steady average is the signature of a stable climate, the opposite of rapid change.

11. (C) Ranges are Mon $29-18=11$, Tue $26-19=7$, Wed $37-16=21$, Thu $34-28=6$, Fri $30-20=10$. The smallest is Thursday at 6 degrees. Wednesday's 21 is the LARGEST, not the smallest, and the highest maximum is irrelevant to range; Tuesday's 7 is close but still larger than Thursday's 6, and the lowest maximum does not by itself give the smallest range; Monday's 11 is far from the smallest, and being the earliest day is incidental. The smallest range is found only by computing max-minus-min for every day.

12. (D) Rain fell on Tue, Fri, Sat, Sun = 4 days; total = $8+12+36+4 = 60$ mm; average per RAINY day = $60/4 = 15$ mm. The first distractor counts dry days as rainy; the second computes the correct totals but wrongly divides by 7 instead of 4 for 'per rainy day'; the third drops a genuine rainy day and miscounts the total. The phrase 'per rainy day' fixes the divisor at the number of days it actually rained.

13. (A) Daily ranges: $27-17=10$, $31-19=12$, $36-22=14$, $30-18=12$; their average is $(10+12+14+12)/4 = 48/4 = 12$. Averaging the maxima alone gives a temperature, not a range; subtracting the lowest min from the highest max gives one giant span across all days, not the average daily range; and using only two days ignores half the data. The average daily range is the mean of each day's own max-minus-min.

14. (A) The max-min thermometer uses index markers left behind at the highest and lowest temperatures, so it stores both extremes unattended. A reading at one instant cannot reveal the whole day's extremes; the instrument records, it does not need hourly human sampling and averaging; and rainfall has nothing to do with temperature extremes — it is measured by a separate rain gauge.

15. (D) Once the jar overflows, the excess is lost, so the recorded figure is only the minimum that fell — the real total could be far higher. 'Exactly 120' treats a saturated instrument as exact; 'at most 120' is backwards, since the true amount is at least that; and an overflow does not signal that rain stopped — water kept arriving and simply spilled. An instrument can only certify within its capacity.

16. (A) A hygrometer measures humidity (68%); an anemometer measures wind speed (24 km/h). Temperature comes from a thermometer and sunshine from a sunshine recorder, not the hygrometer or anemometer; rainfall comes from a rain gauge, not a hygrometer, and humidity is not measured by an anemometer; and the final option simply swaps the two correct instruments. Each element has its own dedicated tool.

17. (C) Relative humidity falls as air warms because warm air can hold more moisture, so the same vapour is a smaller percentage of its larger capacity — the afternoon heat (which lags the noon Sun) drives the 14:00 dip. The dip need not mean less actual vapour escaped 'upward'; hygrometers do not self-reset on a schedule; and wind does not empty an instrument of its reading. The cause is rising temperature lowering relative humidity.

18. (C) For the same storm, a wide mouth catches more litres than a narrow one, but the DEPTH (volume divided by mouth area) is identical, so depth is mouth-size-independent and comparable everywhere. Depth measures amount, not speed; litres certainly can be measured, depth is just the better invariant; and a wide gauge collects MORE litres, not fewer — the point is that depth removes that dependence entirely.

19. (D) Even past noon, incoming heat exceeds outgoing heat for a while, so stored heat in the ground keeps the air warming until balance is reached near 2 p.m. The Sun is not nearer the Earth at 2 p.m.; a properly working thermometer does not lag by hours; and humidity does not fall to zero at 2 p.m., nor would dry air alone explain a timed peak. The lag comes from the delay between peak sunlight and peak stored-heat release.

20. (B) All night the ground radiates away heat with nothing coming in, so cooling accumulates and the minimum arrives after the longest cooling period — just before dawn. The Earth's distance from the Sun does not cycle within one night; the Moon does not draw heat from the ground; and a sound thermometer does not drift down on its own. The lag of the minimum past midnight comes from continued heat loss without resupply.

21. (A) Same time, different sunlight, different temperature — so sunlight, not the clock, drives the heating. The clock is identical for both and cannot be the cause; cold air does not selectively avoid sunlit ground; and the difference is real, not a fault — two correct thermometers SHOULD differ when one is sunlit and one is shaded. Holding time constant isolates sunlight as the heating agent.

22. (B) Slanting rays spread their energy thin over a larger patch of ground, weakening the heating — this is true both at the poles and at a winter noon, where the Sun sits low. Distance to the Sun barely changes seasonally and is not the cause; winter days are shorter, not longer; and air is not seasonally 'thicker' in a way that blocks sunlight. The shared cause is ray angle, not distance, hours, or air thickness.

23. (A) Uneven solar heating makes warm air rise, creating pressure differences that move air as wind, and it evaporates water that condenses and rains — one energy source, two effects. Rays do not mechanically push air to make wind; the Sun WARMS, not cools, water to evaporate it; and the Moon does not drive weather. The whole engine runs on uneven heating plus evaporation.

24. (D) Direct overhead equatorial rays pack energy into a small area (hot); slanting polar rays spread the same beam over more area (cold) — the per-area intensity is the key. The

two do not receive equal energy per area; long polar summer days cannot overcome the steeply slanting, weak rays; and slanting rays carry LESS energy per area than direct ones, not more, so the final option inverts the physics.

25. (B) The atmosphere is chaotic: small errors amplify, so a specific distant day is unpredictable, but the long-run averaged climate (the season) is stable and forecastable. The Sun does not weaken a year ahead; climate is very much studied, indeed it is what makes the seasonal forecast possible; and thermometers work fine beyond a day — the limit is the growth of uncertainty, not the instruments.

26. (C) Hot deserts are very hot by day with wide swings and minimal rain — City B (46/7, 90 mm) fits. City A's constant heat and 2400 mm of rain is a tropical rainforest, not a desert; City C with sub-freezing temperatures is a COLD/polar type despite low rain; and City D is a moderate, semi-humid place, not a hot desert. Both 'hot' and 'dry' must hold together, which only B satisfies.

27. (B) Steady heat, very high humidity, heavy spread-out rain and a tall canopy are rainforest signatures; grasping limbs suit climbing and loud calls beat poor visibility among leaves. Polar regions are freezing, not 24-29 deg C; deserts are dry, not 2300 mm, and tails do not store water; and snowfields lack a dense warm canopy. Only the rainforest reading ties every clue together.

28. (A) Fat insulates, a compact body has low surface-area-to-volume to retain heat, and white fur camouflages in snow — all cold-climate, heat-conserving traits, so a rainforest would overheat the animal and its white coat would betray it against green. They do not shed heat (the opposite); white is camouflage for snow, not green leaves; and fat is energy and insulation, not a water store. The shared purpose is surviving cold.

29. (C) A compact body has less surface per unit volume, so it radiates heat away more slowly — ideal for the cold; the long, large-tailed body of Y loses heat too quickly. In freezing cold, soaking up faint Sun cannot offset rapid heat loss; shape strongly governs heat loss; and a compact body exposes LESS surface, not more. Surface-area-to-volume is the deciding rule.

30. (D) Wide paws do double duty — distributing weight on snow and paddling in water — while fur and fat together insulate against cold air and water. Fat insulates and stores energy, not primarily for floating to ambush; dense fur conserves warmth rather than cools; and swimming has nothing to do with climbing trees, which polar bears do not do. Each feature's main job is heat conservation or movement on ice and water.

31. (D) The white coat camouflages the bear against snow so it can approach prey unseen; it is not a heat-absorber. White surfaces reflect rather than absorb the most heat, so the first option is physically wrong; the coat is not for being spotted by other bears; and

camouflage exists precisely so prey does NOT see the bear coming. Its job is concealment in a snowy climate.

32. (C) Packed together, each bird shields the others, cutting the surface exposed to wind and cold, so the group conserves heat. The aim is to lose less heat, not shed excess; huddling on ice has nothing to do with herding fish; and trapping COLD air to cool themselves would be self-defeating in freezing weather — they conserve warmth, not invite chill.

33. (C) Ear size tunes heat exchange via surface area: big thin ears radiate heat away in the heat; small ears minimise heat loss in the cold. Hearing is not the climatic point here; large ears in the heat SHED heat, they do not trap it; and ear size tracks climate, not merely body size. Surface area for heat exchange is the unifying rule.

34. (D) Both colourings are camouflage matching the background — green leaves, white snow — to hide from predators or prey. Colour here is about concealment, not warming the body; neither animal glows in the dark; and camouflage works by blending in, the opposite of being the brightest, most startling colour. The shared principle is background-matching.

35. (C) Grasping limbs, a balancing tail and loud calls are all rainforest tools — for climbing trees and signalling through foliage — useless on treeless ice. None of these traits aids cold survival; they are not sand-burrowing tools either; and the monkey cannot fly. Its whole body plan is matched to a warm forest, not the polar climate.

36. (D) A warm, wet, year-round-stable climate offers plentiful, diverse food and shelter, so many species can live together — that is why rainforests are so biodiverse. Cold does not speed reproduction; rainforests are densely PLANTED, which is the basis of their abundance, not bare of plants; and biodiversity is set by climate and resources, not by tourist numbers.

37. (B) A slender body and big thin ears maximise surface area, and short fur gives little insulation — all good for losing heat, so a hot climate. Large ears RELEASE heat, they do not hold warmth; a slender body loses heat fast, the opposite of conserving it; and shape and ear size strongly affect heat loss. The whole plan is built for shedding heat.

38. (B) Water heats and cools slowly and supplies moisture, so a coast has a gentle range and high humidity, while the inland site, lacking that buffer, swings widely and stays dry. Altitude is not stated and is a separate factor; being inland does not bring a place closer to the Sun; and the contrast is a year-long climate difference, not an artefact of which days were sampled.

39. (A) Big day-to-day jumps in sunshine over six days are weather variability, not a climate shift — six days cannot show a climate change. The figures are obviously not

constant; and they are clearly NOT the same each day, so cloud cover plainly varied. The lesson is the rapid changeability of weather over a short span.

40. (D) Scorching days, sharp night cooling, big swings and almost no rain are hot-desert signatures — clear dry air heats and cools fast. Heat does not imply wet, so it is not a rainforest; polar climates are persistently cold, not 48 deg C by day; and a coast would show a SMALL range and more humidity, the opposite of wide swings and dryness.

41. (D) Insulating fat suits cold but overheats a rainforest animal; grasping limbs suit forest trees but find nothing to grip on open ice — each is a help in one climate, a hindrance in the other. A grasping tail does not grip ice for warmth; fat is not rain-cushioning; and both features are very much climate-dependent, not universal.

42. (C) Low surface-area-to-volume conserves heat (compact polar body); high surface-area-to-volume sheds heat (lean, big-eared desert body) — one rule, opposite directions. The first option inverts both: compact bodies lose heat SLOWER, lean ones FASTER; shape is about heat, not just apparent size; and the lean-versus-round contrast is driven by climate heat needs, not food abundance.

43. (B) Migration tracks the predictable seasonal climate — escaping the northern winter, returning as it warms — not the weather of one chosen day. The same individuals return year after year, so they do not die out and respawn; and animals do not transform into different species with the season. The recurring seasonal cycle (climate) governs the journey, not a single day's weather.

44. (D) The repeating seasonal climate cycle drives the yearly round trip, and birds usually leave on cues like shortening days and dwindling food, departing ahead of the worst cold. A single day's random weather does not govern a yearly cycle; the birds do not settle permanently — they return each spring; and the cue is a broad seasonal signal, not one tree's leaves nor a never-repeating event.

45. (D) Depth equals collected volume divided by the funnel's mouth area; a wide funnel catches more volume over a larger area, a narrow one less over a smaller area, and the depths come out equal. So the wide gauge does not read deeper; the narrow gauge does not either; and far from being incomparable, equal depths are precisely what makes cross-station comparison fair — that is the whole point of using depth.

46. (C) Climate (the multi-year averaged wettest month) is predictable, while one specific distant day is not, because small uncertainties amplify in the chaotic atmosphere. The Sun is not selectively unpredictable on certain days; rainfall does not obey the calendar like a rule with random exceptions; and the limitation is the growth of uncertainty, not faulty instruments. Averaged pattern is forecastable; a single far-off day is not.

47. (D) Place 1 hovers near 26-27 (small swings) while Place 2 leaps between 16 and 45 (large swings); a shared average conceals this real climatic difference in variability. A

matching mean does not make climates identical; high variability is itself a climate feature, not an absence of climate; and a steady place still has daily weather — small swings are not zero swings.

48. (B) Humidity (71%) is precisely the water-vapour content of the air. Rainfall of 0 mm only says no rain fell, not how much vapour is present; a high temperature raises the air's CAPACITY for vapour but does not by itself state the amount held; and wind speed measures air motion, not its moisture. Humidity is the dedicated vapour measure.

49. (B) Freezing temperatures, snow and a very long night signal a polar climate; warm-and-wet with a dense forest signals tropical rainforest. A wide range does not make the freezing report a hot desert, and precipitation alone does not make the second polar; precipitation does not make both tropical (snow is polar); and a max just above zero with a 19-hour night is polar, not a temperate coast, while a warm forest is rainforest, not desert.

50. (A) A single cold day is weather; a polar climate requires year-round cold, so one freezing day proves nothing. Cold days occur in many non-polar places, so the first option is false; one cold day cannot establish a climate, so the child is not right; and temperature is central to defining a polar region, so the last option is wrong. Weather versus climate is the key distinction.

51. (A) A heritable colour that consistently matches the leafy background across generations and improves survival is an adaptation, not chance. Green colour does not warm the frog; it is not a desert reflectance trait; and frogs do not glow to lure prey in the dark. The reliable, inherited background-match with a survival benefit is the hallmark of adaptation.

52. (A) An adaptation is heritable, long-term and climate-matched — polar bears inheriting fur and fat suited to a cold climate fit exactly. Feeling warm one afternoon is a momentary experience, not a heritable feature; learning to open fruit is a one-time learned behaviour, not inherited; and flying indoors once is a single act, not a heritable trait. Only the inherited, climate-fitting feature qualifies.

53. (D) A light reaching beak lets the toucan feed at the tips of thin branches; thick fur insulates the bear in cold. A beak is not a fat store, and fur conserves warmth rather than cools; toucans do not dig burrows, and the bear lives among ice, not green leaves; and a beak does not heat the head, nor does fur improve swimming speed. Each feature's real job is feeding and insulation respectively.

54. (B) Four months (Jun-Sep) carry the bulk of the rain while the rest are dry — a strongly seasonal monsoon-type pattern, not even spread. A large yearly total alone does not make it a rainforest, since rainforest rain is spread through the year, not crammed into four months; small monthly figures in a warm region do not make it polar; and 'at least

some rain each month' does not mean evenly spread, since June's 260 dwarfs March's 4. The defining feature is concentration into a season.

55. (A) The within-day variation is the range = max - min = 33 - 19 = 14 deg C. The maximum alone is just the peak, not the variation; the minimum alone is just the floor; and adding the two gives a meaningless 52, not a variation. Variation within a day is the difference between the extremes, the range.

56. (C) Direct equatorial sunlight heats and drives evaporation feeding wet rainforests, while slanting polar sunlight leaves the poles cold and icy — the three facts cohere. The poles are cold, not hot, and ice forms there, not at the equator; the equator is hot, not cold, and rainforests need warmth and rain; and the placement of rainforests and ice caps is clearly tied to sunlight intensity, not accidental.

57. (B) For a while past noon, incoming heat still exceeds heat lost, so the ground and the air above keep warming until balance, producing the afternoon peak. The Sun is overhead at noon, not mid-afternoon; air does not 'travel down from the Sun'; and clouds do not reliably gather at noon every day. The lag is a heat-storage effect, not a Sun-position or cloud effect.

58. (B) Solar heating evaporates water faster where it is warmer, so the sunlit puddle vanishes first — and that same Sun-driven evaporation is the first step in cloud and rain formation. The shaded puddle did not selectively sink while the other was pushed sideways; sunlight does not have weight that presses water out; and the puddles plainly lost different amounts, since one is dry and one is wet. Evaporation rate rises with the Sun's heat.

59. (B) A genuine coast has high humidity and a small range because the sea adds moisture and damps temperature swings; L's low humidity and wide range mark it as inland, contradicting the 'coastal' label. Nearness to the sea strongly affects both humidity and range, so the first option is false; mountains behind L block sea moisture rather than add it; and the sea supplies moisture rather than removing it, so the last option inverts the physics.

60. (D) Weather is short-term, climate is the multi-year average, and adaptations are inherited features tuned to the long-term climate — all three correct. The first option swaps weather and climate and wrongly ties adaptation to single days; the second wrongly equates weather and climate and reduces adaptation to a day's learning; and the third falsely claims climate is unmeasurable and severs adaptation from climate. Only the precise definitions hold together.